Quizrr

India's Most Comprehensive & the Most Relevant Test Series designed according to the latest pattern of exams!

JEE MAIN | JEE ADV. | BITSAT | WBJEE | MHT CET

and many more...

Click here to join Test Series for 2022

It's time for you to crack upcoming IIT JEE Main & Advanced and other competitive exams with India's Most Trusted Online Test Series. **Many questions at JEE Main 2021 were same/similar to the ones asked in our test series.** That's the power of our test series!

Trusted by thousands of students & their parents across the nation

Our result in JEE Main 2021

150+
Got **99+ percentile** (overall)

301
Got **99+ percentile** in one or more subjects

85%
Improved their score by **25 percentile**

89%
Felt **overall confident** after the test series

Click here to join Test Series for 2022

MARKS

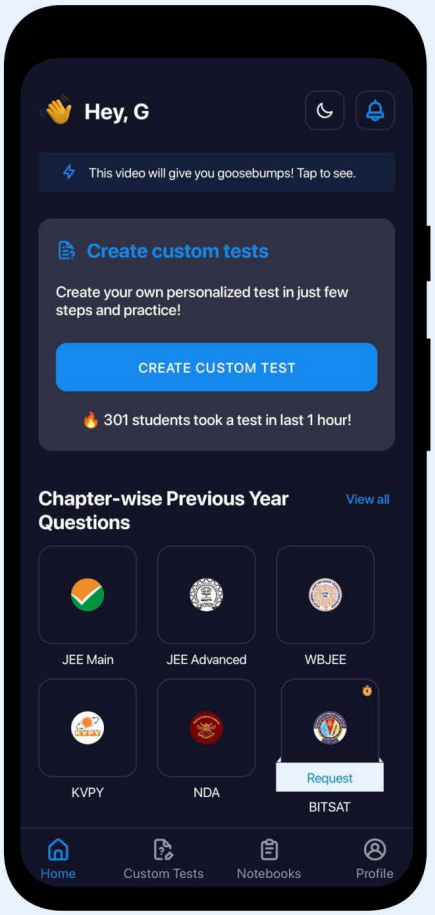
FREE Question Bank & Previous Year Questions for

JEE MAIN | JEE ADV. | BITSAT | WBJEE | MHT CET

and many more...

GET IT ON Google Play

Click on this button



😎 Why download MARKS?

- 📚 Chapter-wise PYQ of JEE Main, JEE Advanced, NEET, AIIMS, BITSAT, WBJEE, MHT CET etc.
- 📚 Chapter-wise NTA Abhyas questions
- 🎯 Daily practice challenge and goal completion
- 🔖 Bookmark important questions and add them to your notebooks
- ✍️ Create unlimited Custom Tests

And all this for FREE. Yes, FREE! So what are you waiting for, download MARKS now.

4.8	30,000+	1,00,000+
Rating on Google Play	Students using daily	Questions available

GET IT ON Google Play

Click on this button

HYPERBOLA

HYPERBOLA

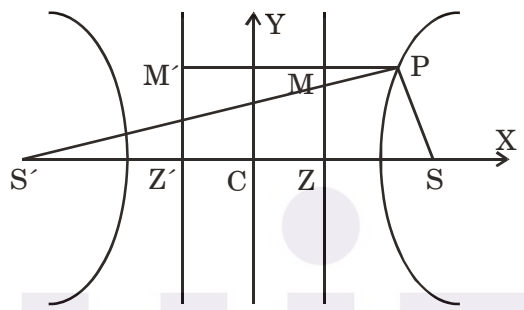
Definition

A Hyperbola is the locus of a point which moves in a plane so that the ratio of its distance from a fixed point (called focus) and a fixed line (called directrix) is a constant which is greater than one, this ratio is called eccentricity and is denoted by e. For hyperbola $e > 1$.

Let S be the focus, QN be the directrix and P be any point on the hyperbola. Then by definition

$$\frac{PS}{PN} \text{ or } PS = e\ PN,\ e > 1$$

where PN is the length of the perpendicular from P on the directrix QN. In other words a hyperbola is the locus of a point which moves in such a way that the difference of its distance from two fixed points (called foci) is constant.



PM and PM´ are perpendicular to the directrices MZ and M´Z´.

$PS' - PS = e\ (PM' - PM)$

$eMM' = e\ (2a/e) = 2a$ = constant.

Standard Equations of Hyperbola

Let S be the focus and ZM the directrix of a hyperbola. Since $e > 1$, we can divide SZ internally and externally in the ratio e : 1, let the points of division be A and A´ as in the figure.

Let AA´ = 2a and is bisected at C.

Then, $SA = e.\ AZ$, $SA' = e.ZA'$

$\Rightarrow \quad SA + SA' = e\ (AZ + ZA') = 2ae$

i.e. $\quad 2SC = 2ae$ or $SC = ae$

similarly by subtraction,

$$SA' - SA = e\ (ZA' - ZA) = 2e.ZC$$

$\Rightarrow \quad 2a = 2eZC \Rightarrow ZC = a/\ e$

Now, take C as the origin, CA as the x-axis, and the perpendicular line CY as the y-axis. Then, S is the point (ae, 0) and ZM the line $x = a/\ e$. Let P (x, y) be any point on the hyperbola.

Then the condition $PS^2 = e^2$. (distance of P from ZM)^2 gives $(x - ae)^2 + y^2 = e^2\ (x - a/\ e)^2$

or $\quad x^2(1-e)^2 + y^2 = a^2(1-e)^2$ i.e., $\dfrac{x^2}{a^2} - \dfrac{y^2}{a^2(e^2-1)} = 1$ ...(i)

Since $e > 1$, $e^2 - 1$ is positive. Let $a^2(e^2-1) = b^2$. Then the equation (i) becomes $\dfrac{x^2}{a^2} - \dfrac{y^2}{b^2} = 1$.

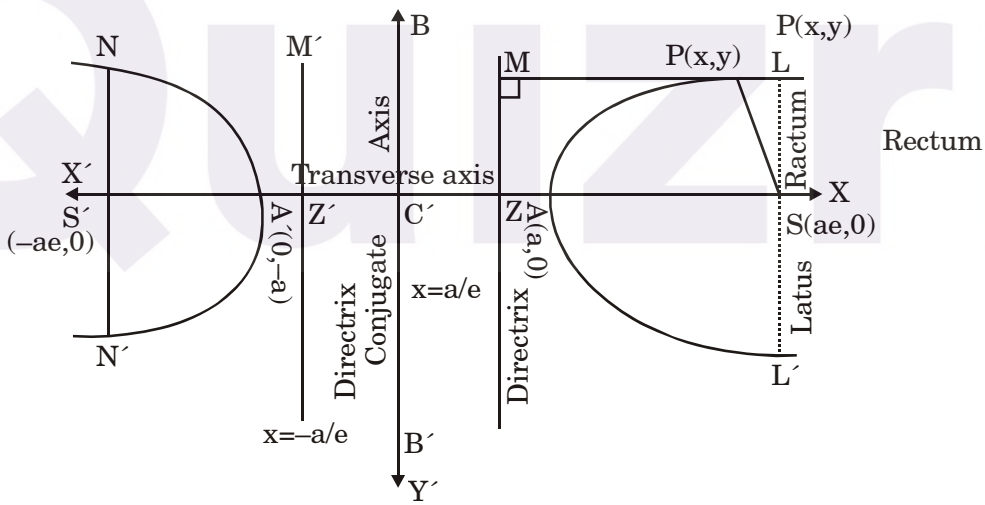
The eccentricity e of the hyperbola $\dfrac{x^2}{a^2} - \dfrac{y^2}{b^2} = 1$ is given by the relation $e^2 = \left(1 + \dfrac{b^2}{a^2}\right)$.

Since the curve is symmetrical about the y-axis, it is clear that there exists another focus S´ at (– ae, 0) and a corresponding directrix Z´M´ with the equation x = – a/e, such that the same hyperbola is described if a point moves so that its distance from S´ is e times its distance from Z´M´.

Terms Related to Hyperbola

Symmetry : Since only even powers of x and y occur in the above equation, so the curve is symmetrical about both the axes.

Foci : S and S´ are the two foci of the hyperbola and their coordinates are (ae, 0) and (– ae, 0) respectively, then distance between foci is given by SS´ = 2ae.



Directries : ZM and Z´M´ are the two directrices of the hyperbola and their equations are $x = \dfrac{a}{e}$ and $x = -\dfrac{a}{e}$ respectively, then the distance between directries is given by ZZ´ = $\dfrac{2a}{e}$

Axes : The lines AA´ and BB´ are caled the transverse axis and conjugate axis respectively of the hyperbola. The length of transverse axis = AA´ = 2a. The length of conjugate axis = BB´ = 2b.

Centre : The point of intersection C of the axes of hyperbola is called the centre of the hyperbola. All chords passing through C, are bisected at C.

Vertices : The point A ≡ (a, 0) and A´ ≡ (– a, 0) where the curve meets the line joining the foci S and S´ are called the vertices of the hyperbola.

Focal chord : A chord of the hyperbola passing through its focus is called a focal chord.

Focal distance : The focal distance of any point (x, y) on the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ are ex – a and ex + a.

The difference of the focal distance of any point on the hyperbola is constant and equal to the length ofo the transverse axis of the hyperbola. If P is any hyperbola, then

S´P – SP = 2a = Transverse axis.

Latus Rectum : If LL´ and NN´ are the latus rectum of the hyperbola then these lines are perpendicular to the transverse axis AA´, passing through the foci S and S´ respectively.

$$L \equiv \left(ae, \frac{b^2}{a}\right), \qquad L' \equiv \left(ae, -\frac{b^2}{a}\right), \quad N \equiv \left(-ae, \frac{b^2}{a}\right) \quad N' \equiv \left(-ae, \frac{-b^2}{a}\right)$$

Length of latus rectum = $LL' = \frac{2b^2}{a} = NN'$

Eccentricity of the Hyperola

We know that SP = ePM or $SP^2 = e^2\, PM^2$

or $(x - ae)^2 + (y - 0)^2 = e^2\left(x - \frac{a}{e}\right)^2$ or $(x - ae)^2 + y^2 = (ex - a)^2$

or $x^2 + a^2e^2 - 2aex + y^2 = e^2\, x^2 - 2aex + a^2$ or $x^2\,(e^2 - 1) - y^2 = a^2\,(e^2 - 1)$

$$\frac{x^2}{a^2} - \frac{y^2}{a^2(e^2 - 1)} = 1$$

On comparing with $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$, we get $b^2 = a^2\,(e^2 - 1)$ or $e = \sqrt{1 + \frac{b^2}{a^2}}$

or $$e = \sqrt{\mathbf{1 + \frac{(conjugate\ axis)^2}{(transverse\ axis)^2}}}$$

General Equation of Hyperbola

The equation of hyperbola, whose focus is pont (h, k) directrix is lx + my + n = 0 & ecentricity 'e' is given by $(x - h)^2 + (y - k)^2 = \frac{e^2(lx + my + n)^2}{(l^2 + m^2)}$

Illustration 1

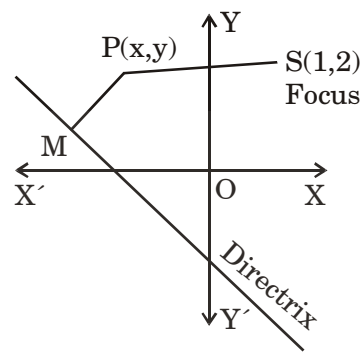
Find the equation of the hyperbola whose focus is (1, 2) directrix is the line $x + y + 1 = 0$ and eccentricity $\frac{3}{2}$.

Solution :

Let S (1, 2) be the focus and let P (x, y) be a point on the hyperbola. Draw perpendicular PM from P on the directrix $x + y + 1 = 0$. Then

SP = e PM

$$\Rightarrow \quad \sqrt{(x-1)^2 + (y-2)^2} = \frac{3}{2}\left(\frac{x+y+1}{\sqrt{1^2+1^2}}\right)$$



$$\Rightarrow \quad (x-1) + (y-2)^2 = \frac{9}{4}\left[\frac{(x+y+1)^2}{2}\right]$$

$$\Rightarrow \quad 8\,[(x-1)^2 + (y-2)^2] = 9\,[(x+y+1)^2]$$

$$\Rightarrow \quad 8x^2 + 8y^2 - 16x - 32y + 40$$

$$= 9x^2 + 9y^2 + 9 + 18xy + 18x + 18y$$

$$\Rightarrow \quad x^2 + y^2 + 18xy + 34x + 50y - 31 = 0$$

This is the equation of the required hyperbola.

Illustration 2

Find the eccentricity of the conic represented by $x^2 - y^2 - 4x + 4y + 16 = 0$

Solution :

We have $x^2 - y^2 - 4x + 4y + 16 = 0$

$$\Rightarrow \quad (x^2 - 4x) - (y^2 - 4y) = -16 \quad \Rightarrow \quad (x^2 - 4x + 4) - (y^2 - 4y + 4) = -16$$

$$\Rightarrow \quad (x-2)^2 - (y-2)^2 = -16 \quad \Rightarrow \quad \frac{(x-2)^2}{4^2} - \frac{(y-2)^2}{4^2} = -1$$

Shifting the origin at (2, 2), we obtain $\frac{X^2}{4^2} - \frac{Y^2}{4^2} = -1$, where $x = X + 2$, $y = Y + 2$

This is rectangular hyperbola, whose eccentricity is always $\sqrt{2}$

Illustration 3

Find the centre, eccentricity, foci, directrices and the lengths of the transverse and conjugate axes of the hyperbola, whose equation is $(x-1)^2 - 2(y-2)^2 + 6 = 0$

Solution :

The equation of the hyperbola can be written as $(x - 1)^2 - 2(y - 2)^2 + 6 = 0$

or $-\dfrac{(x-1)^2}{\sqrt{(6)}^2} + \dfrac{(y-2)^2}{(\sqrt{3})^2} = 1$ or $\dfrac{Y^2}{(\sqrt{3})^2} - \dfrac{X^2}{(\sqrt{6})^2} = 1$

Where $Y = (y - 2)$ and $X = (x - 1)$ ...(1)

$\therefore$ centre : $X = 0$, $Y = 0$ i.e., $(x - 1) = 0$, $x = 1$ & $(y - 2) = 0$, $y = 2$.

So, $a = \sqrt{3}$ and $b = \sqrt{6}$ so transverse axis = $2\sqrt{3}$, and conjugate axis = $2\sqrt{6}$.

Also $b^2 = a^2 (e^2 - 1)$ $\Rightarrow$ $6 = 3 (e^2 - 1)$ i.e., $e = \sqrt{3}$

In (X, Y) coordinates, foci are $(0, \pm ae)$

i.e., $(0, \pm 3)$ $\therefore$ foci are $(1, 2 \pm 3)$ i.e., $(1, 5)$ and $(1, -1)$

Equations of directrices, $Y = \pm a/e$

$\therefore$ directrices $y - 2 = \pm \sqrt{3}/\sqrt{3} = \pm 1$ or $y = 3$, $y = 1$.

Illustration 4

Find the equation of the hyperbola whose foci are (8, 3) (0, 3) and eccentricity = $\dfrac{4}{3}$.

Solution :

The centre of the hyperbola is the mid-point of the line joining the two foci. So the coordinates of the centre are $\left(\dfrac{8+0}{2}, \dfrac{3+3}{2}\right)$ i.e. (4, 3).

Let 2a and 2b be the length of transverse and conjugate axes and let e be the eccentricity. Then the equation of the hyperbola is $\dfrac{(x-4)^2}{a^2} - \dfrac{(y-3)^2}{b^2} = 1$ ...(i)

Now, distance between the two foci = 2ae

$\Rightarrow$ $\sqrt{(8-0)^2 + (3-3)^2} = 2ae \Rightarrow$ $ae = 4$ $\Rightarrow$ $a = 3$ $\left(\because e = \dfrac{4}{3}\right)$

Now, $b^2 = a^2 (e^2 - 1)$ $\Rightarrow$ $b^2 = 9\left(-1 + \dfrac{16}{9}\right) = 7$

Thus, the equation of the hyperbola is

$\dfrac{(x-4)^2}{9} - \dfrac{(y-3)^2}{7} = 1$ [Putting the values of a and b in (i)]

$\Rightarrow$ $7x^2 - 9y^2 - 56x + 54y - 32 = 0$

Parametric Equations of the Hyperbola

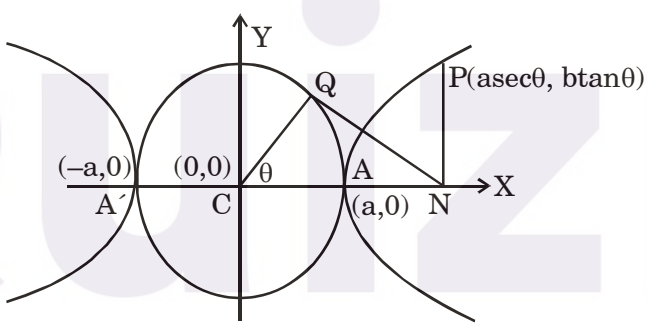
Since coordinate x = a secθ and y = b tanθ satisfy the equation $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ for all real values of θ therfore, x = a secθ, y = b tanθ are the parametric equations of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$, where the parameter $0 \le \theta < 2\pi$

Hence, the coordinates of any point on the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ may be taken as (a secθ, b tanθ). This point is also called the point θ. The angle θ is called the eccentric angle of the point (a secθ, b tanθ) on the hyperbola.

Auxiliary Circle

A circle drawn with the centre C and T.A. as a diameter is called the Auxiliary Circle of the hyperbola. Equation of the auxiliary circle is $x^2 + y^2 = a^2$.

Note from the figure that P and Q are called the "Corresponding Points" on the hyperbola and the auxiliary circle. 'θ is called the eccentric angle of the point 'P' on the hyperbola ($0 \le \theta < 2\pi$).

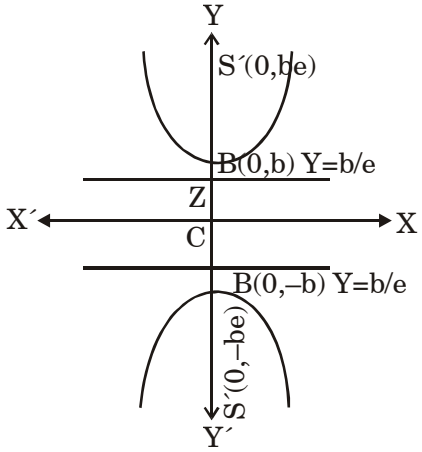


Conjugate Hyperbola

The hyperbola whose transverse and conjugate axes are respectively the conjugate and transverse axes of a given hyperbola is called the conjugate hyperbola of the given hyperbola.

The conjugate hyperbola of the hyperbola

$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ is $\frac{x^2}{a^2} - \frac{y^2}{b^2} = -1$.



Properties of Hyperbola and its Conjugate

	Hyperbola	**Conjugate Hyperbola**
Standard equation	$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$	$\frac{-x^2}{a^2} + \frac{y^2}{b^2} = 1$ or $\frac{x^2}{a^2} - \frac{y^2}{b^2} = -1$
Centre	(0, 0)	(0, 0)
Eq. of transverse axis	y = 0	x = 0
Eq. of conjugate axis	x = 0	y = 0
Length of transverse axis	2a	2b
Lentgh of conjugate axis	2b	2a
Foci	(± ae, 0)	(0, ± be)
Equation of directrices	x = ± a/e	y = ± b/e
Vertices	(± a, 0)	(0, ± b)
Eccentricity	$e = \sqrt{\frac{a^2 + b^2}{a^2}}$	$e = \sqrt{\frac{a^2 + b^2}{b^2}}$
Length of latus rectum	$2b^2/a$	$2a^2/b$
Parameter Coordinates	(a secθ, b tanθ)	(b secθ, a tanθ)
Focal radii	$SP = ex_1 - a$ and $S'P = ex_1 + a$	$SP = ey_1 - b$ and $S'P = ey_1 + b$
Difference of focal radii (S´P – SP)	2a	2b
Tangent of the vertices	x = ± a	y = ± b

Position of A Point P w.r.t. Hyperbola :

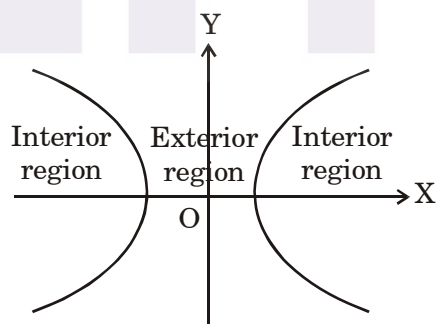
Let S = 0 be the hyperbola and P (x_1, y_1) be the point and $S_1 \equiv S(x_1, y_1)$.

Then

$S_1 < 0 \Rightarrow$ P is in the exterior region

$S_1 > 0 \Rightarrow$ P is in the interior region

$S_1 = 0 \Rightarrow$ P lies on the hyperbola



Important :

Note that most of the results proved for hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ can be obtained from the same formula for ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, by replacing b^2 by $-b^2$ in conditions for ellipse.

Illustration 5

Show that the equation $x^2 - 2y^2 - 2x + 8y - 1 = 0$ represents a hyperbola. Find the coordinates of the centre, lengths of the axes, eccentricity, latus rectum, coordinates of the foci and vertices and equations of directrices of the hyperbola.

Solution :

$x^2 - 2y^2 - 2x + 8y - 1 = 0$

$\Rightarrow \quad (x^2 - 2x) - 2(y^2 - 4y) = 1$

$\Rightarrow \quad (x^2 - 2x + 1) - 2(y^2 - 4y + 4) = -6$

$\Rightarrow \quad (x-1)^2 - 2(y-2)^2 = -6$

$$\Rightarrow \quad \frac{(x-1)^2}{(\sqrt{6})^2} - \frac{(y-2)^2}{(\sqrt{3})} = -1 \quad ...(i)$$

Shifting the origin at (1, 2) without rotating the coordinates axes and denoting the new coordinates with respect to these axes by X and Y, we have

$x = X + 1$ and $y = Y + 2$ ...(ii)

Using these relations, equation (i) reduces to

$$\frac{X^2}{(\sqrt{6})^2} - \frac{Y^2}{(\sqrt{3})^2} = -1 \quad ...(iii)$$

This equation is of the form $\frac{X^2}{a^2} - \frac{Y^2}{b^2} = -1$, where $a^2 = (\sqrt{6})^2$ and $b^2 = (\sqrt{3})^2$. So, we have :

Centre The coordinates of the centre with respect to the new axes are (X = 0, Y = 0).

So, the coordinates of the centre with respect to the old axes are (1, 2) [Putting X = 0, Y = 0 in (ii)]

Lengths of the Axes : Since the transverse axis of the hyperbola is along new Y-axis.

$\therefore$ Transverse axis = 2b = $2\sqrt{3}$, Conjugate axis = 2a = $2\sqrt{6}$.

Eccentricity : The eccentricity e is given by $e = \sqrt{1 + \frac{a^2}{b^2}} = \sqrt{1 + \frac{6}{3}} = \sqrt{3}$

Latus rectum : Length of the latus rectum = $\frac{2a^2}{b} = \frac{12}{\sqrt{3}} = 4\sqrt{3}$.

Foci : The coordinates of foci with respect to the new axes are

(X = 0, Y = ± be) i.e. (X = 0, Y = ± 3).

So, the coordinates of the vertices with respect to the old axes are (1, 2 ± 3) i.e. (1, 5) and (1, – 1) [Putting X = 0, Y = ± 3 in (ii)]

Vertices : The coordinates of the vertices with respect to the new axes are X = 0, Y = ± b i.e. (X = 0, Y = ± $\sqrt{3}$)

So the coordinates of the vertices w.r.t. to the old axes are

(1, 2 ± $\sqrt{3}$) i.e. (1, 2 + $\sqrt{3}$) and (1, 2 − $\sqrt{3}$) [Putting X = 0, Y = + $\sqrt{3}$ in (ii)]

Directrices : The equations of the directrices with respect to the new axes are Y = ± b/e i.e. Y = ± 1

So the equations of the directrices with respect to the old axes are

y = 2 ± 1 i.e. y = 1 and y = 3 [Putting Y = ± 2 in (ii)]

Illustration 6

If e and e′ be the eccentricities of a hyperbola and its conjugate, prove that $\frac{1}{e^2}+\frac{1}{e^2}=1$.

Solution :

Let the equation of the hyperbola be $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$ ...(i)

Then the equation of the hyperbola conjugate to (i) is $\frac{x^2}{a^2}-\frac{y^2}{b^2}=-1$ ...(ii)

$$e = \text{Eccentricity of (i)} = \sqrt{1+\left(\frac{\text{conjugate axis}}{\text{Transverse axis}}\right)^2}$$

$$\Rightarrow \quad e=\sqrt{1+\left(\frac{2b}{2a}\right)^2} \quad \Rightarrow \quad e^2=1+\frac{b^2}{a^2} \quad \Rightarrow \quad e^2=\frac{a^2+b^2}{a^2} \qquad ...(iii)$$

$$\text{And, } e' = \text{Eccentricity of (ii)} = \sqrt{1+\left(\frac{\text{conjugate axis}}{\text{Transverse axis}}\right)^2}$$

$$\Rightarrow \quad e'=\sqrt{1+\left(\frac{2a}{2b}\right)^2} \quad \Rightarrow \quad e'^2=1+\frac{a^2}{b^2} \quad \Rightarrow \quad e'^2=\frac{a^2+b^2}{b^2} \qquad ...(iv)$$

From (iii) and (iv), we have

$$\frac{1}{e^2}+\frac{1}{e'^2}=\frac{a^2}{a^2+b^2}+\frac{b^2}{a^2+b^2}=\frac{a^2+b^2}{a^2+b^2} \quad \Rightarrow \quad \frac{1}{e^2}+\frac{1}{e'^2}=1$$

Tangent to the Hyperbola :

1. **Condition for tangency and points of contact :** The condition for the line y = mx + c to be a tangent to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ is that $c^2 = a^2m^2 - b^2$ and the coordinates of the points of contact are $\left(\pm\frac{a^2m}{\sqrt{a^2m^2 - b^2}}, \pm\frac{b^2}{\sqrt{a^2m^2 - b^2}}\right)$

2. **<u>Equation of tangent</u>**

   (a) **Point form :** The equation of the tangent to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ at the point (x_1, y_1) is $\boxed{\frac{xx_1}{a^2} - \frac{yy_1}{b^2} = 1}$

   The equation of tangent at (x_1, y_1) can also be obtained by replacing x^2 by xx_1, y^2 yy_1, x by $\frac{x + x_1}{2}$ y by $\frac{y + y_1}{2}$ and xy by $\frac{xy_1 + x_1y}{2}$. This method is used only when the equation of hyperbola is a polynomial of second degree in x and y.

   (b) **Parametric Form :** The eqn. of the tangent to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ at the point (a secθ b, tanθ) is $\boxed{\frac{x}{a}\sec\theta - \frac{y}{b}\tan\theta = 1}$

   (c) **Slope Form :** The equation of tangent to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = -1$ in terms of slope 'm' is $y = mx \pm \sqrt{a^2m^2 - b^2}$ The coordinates of the points of contact are $\left(\pm\frac{a^2m}{\sqrt{a^2m^2 - b^2}}, \pm\frac{b^2}{\sqrt{a^2m^2 - b^2}}\right)$

3. **Number of Tangents From a Point :** Two tangents can be drawn from a point to a hyperbola. The two tangents are real and distinct or coincident of imaginary according as the given point lies outside, on or inside the hyperbola.

4. **Director Circle :** It is the locus of point from which ⊥ tangents are drawn to the hyperbola. The equation of director circle of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ is $x^2 + y^2 = a^2 - b^2$

5. Equation of the pair of tangents : The equation of the pair of tangents drawn from a point P (x_1, y_1) to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ is $\mathbf{SS_1 = T^2}$

where $S \equiv \frac{x^2}{a^2} - \frac{y^2}{b^2} - 1$, $S \equiv \frac{x_1^2}{a^2} - \frac{y_1^2}{b^2} - 1$ and $T \equiv \frac{xx_1}{a^2} - \frac{yy_1}{b^2} - 1$

Illustration 7

Find the condition for the line $x \cos \alpha + y \sin \alpha = p$ to be a tangent to the hyperobla $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$.

Solution :

We have,

$x \cos \alpha + y \sin \alpha = p \Rightarrow y = -x \cot \alpha - p \operatorname{cosec} \alpha$

$\Rightarrow y = (-\cot \alpha)\, x + (-p \operatorname{cosec} \alpha)$ ...(i)

This will touch $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$, if

$(-p \operatorname{cosec} \alpha)^2 = a^2 (-\cot \alpha)^2 - b^2$ [Using $c^2 = a^2m^2 - b^2$]

$\Rightarrow p^2 \operatorname{cosec}^2 \alpha = a^2 \cot^2 \alpha - b^2 \Rightarrow p^2 = a^2 \cos^2 \alpha - b^2 \sin^2 \alpha$

Illustration 8

Prove that the product of the lengths of the perpendiculars drawn from foci on any tangent to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ is b^2.

Solution :

The equation of any tangent to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ is $y = mx + \sqrt{a^2m^2 - b^2}$ ...(i)

Let S (ae, 0) and S´ (– ae, 0) be two foci of the hyperbola.

Let p and and p´ be the lengths of perpendicular from S (ae, 0) and S´ (– ae, 0) on (i). Then,

p = Length of the ⊥ from S(ae, 0) on (i)

$$= \left|\frac{mae - 0 + \sqrt{a^2m^2 - b^2}}{\sqrt{m^2 + 1}}\right| \left|\frac{mae + \sqrt{a^2m^2 - b^2}}{\sqrt{1 + m^2}}\right|$$

$p' =$ Length of the $\perp$ from $S'(-ae, 0)$ on (i)

$$= \left|\frac{-mae + \sqrt{a^2m^2 - b^2}}{\sqrt{m^2+1}}\right| = \left|\frac{-mae + \sqrt{a^2m^2 - b^2}}{\sqrt{1+m^2}}\right|$$

Now,

$$p.p' = \left|\frac{mae + \sqrt{a^2m^2 - b^2}}{\sqrt{1+m^2}}\right| . \left|\frac{-mae + \sqrt{a^2m^2 - b^2}}{\sqrt{1+m^2}}\right|$$

$$= \left|\frac{-m^2a^2e^2 + a^2m^2 - b^2}{1+m^2}\right| = \left|\frac{-m^2a^2(e^2-1) - b^2}{m^2+1}\right|$$

$$= \left|\frac{m^2b - b^2}{m^2+1}\right| = \frac{(m^2+1)b^2}{(m^2+1)} = b^2$$

Illustration 9

If the tangent at the point (p, q) on the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ cuts the auxiliary circle in points whose ordinates are y_1 and y_2 then q is Harmonic mean of y_1 and y_2.

Solution : Here we choose the tangent as

$$\frac{xp}{a^2} - \frac{yq}{b^2} = 1 \text{ where } \frac{p^2}{a^2} - \frac{q^2}{b^2} = 1 \qquad ...(1)$$

Its intersection with $x^2 + y^2 = a^2$ is given by eliminating x as we are concerned with ordinates

$$\left(1 + \frac{yq}{b^2}\right)^2 . \frac{a^4}{p^2} + y^2 = a^2$$

or $(b^2 + yq)^2 a^4 + b^4y^2p^2 = a^2p^2b^4$

or $y^2(a^4q^2 + b^4p^2) + 2yqb^2a^4 + a^4b^4 - a^2p^2b^4 = 0 \qquad ...(2)$

Above is a quadratic in y. We have to prove that q is H.M. 'H' of y_1 and y_2

$$\text{Now } H = \frac{2y_1y_2}{y_1+y_2} \frac{(a^4b^4 - a^2p^2b^4)}{-2qb^2a^4} \text{ by (2)}$$

$$= \frac{a^4b^4(1 - p^2/a^2)}{-2qb^2.a^4} = \frac{b^2}{-q}\left[\frac{q^2}{b^2}\right] = q, \text{ by (1).}$$

$\therefore$ q is H.M. of y_1 and y_2.

Illustration 10

Determine the equations of common tangents to the hyperbolas $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ and $\frac{y^2}{a^2} - \frac{x^2}{b^2} = 1$.

Solution :

Tangent to $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ is

$$y = m_1 x \pm \sqrt{(a^2 m_1^2 - b^2)} \qquad ...(1)$$

The other hyperbola is $\frac{x^2}{(-b^2)} - \frac{y^2}{(-a^2)} = 1$

Any tangent to it is $y = m_2 x \pm \sqrt{(-b^2) m_2^2 - (a^2)}$ ...(2)

If (1) and (2) are same, then

$m_1 = m_2$ and $a^2 m_1^2 - b^2 = -b^2 m_2^2 + a^2$

or $a^2 m_1^2 - b^2 = a^2 - b^2 m_1^2$

or $(a^2 + b^2)\ m_1^2 = a^2 + b^2$

$\therefore\ m_1^2 = 1 \qquad \therefore\ m_1 = \pm 1$

Hence the common tangents are

$$y = \pm x \pm \sqrt{a^2 - b^2}$$

Equation of Normal In Different Forms :

1. (i) The coordinate of the point of contact are $\left(\pm\frac{a^2 m}{\sqrt{a^2 - b^2 m^2}}, \mp\frac{mb^2}{\sqrt{a^2 - b^2 m^2}}\right)$

2. Equation of normals :

(a) **Point Form :** The equation of the normal to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ at the point (x_1, y_1) is $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ is $\frac{a^2 x}{x_1} + \frac{b^2 y}{y_1} = a^2 + b^2$

(b) Parametric Form : The equation of the normal to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ at the point

(a secθ, b tanθ) is $\frac{ax}{\sec\theta} + \frac{by}{\tan\theta} = a^2 + b^2$

(c) Slope Form : The equation of normal to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ in terms of slope 'm'

is $y = mx \pm \frac{m(a^2 + b^2)}{\sqrt{a^2 - b^2m^2}}$

3. **Number of Normals :** In general, four normals can be drawn to a hyperbola from a point in its plane i.e. there are four points on the hyperbola, the normals at which will pass through a given point. These four points are called the co-normal points.

4. Tangent drawn at any point bisects the angle between the lines joining the point to the foci, where as normal bisects the supplementary angle between the lines.

Illustration 11

Prove that the line lx + my + n = 0 will be a normal to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ if $\frac{a^2}{l^2} - \frac{b^2}{m^2} = \frac{(a^2 + b^2)^2}{n^2}$

Solution :

The equation of the normal at (a sec θ, b tanθ) to the hyperbola

$$a\,x \sin\theta + b\,y = (a^2 + b^2)\tan\theta \qquad ...(i)$$

and the equation of the line is

$$lx + my + n = 0 \qquad ...(ii)$$

Equations (i) and (ii) will represent the same line. If

$$\frac{a\sin\theta}{l} = \frac{b}{m} = \frac{(a^2 + b^2)\tan\theta}{-n}$$

$$\Rightarrow \quad \operatorname{cosec}\theta = \frac{am}{bl} \text{ and } \cot\theta = -\frac{(a^2 + b^2)m}{b^2}$$

$$\therefore \quad \operatorname{cosec}^2\theta - \cot^2\theta = 1 \quad \Rightarrow \quad \frac{a^2m^2}{b^2l^2} - \frac{(a^2 + b^2)m^2}{b^2n^2} = 1$$

$$\Rightarrow \quad \frac{a^2}{l^2} - \frac{(a^2 + b^2)^2}{n^2} = \frac{b^2}{m^2} \quad \Rightarrow \quad \frac{a^2}{l^2} - \frac{b^2}{m^2} = \frac{(a^2 + b^2)^2}{n^2}$$

Illustration 12

If the normal at 'θ' on the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ meets the transverse axis at G, prove that AG.A´G = a^2 ($e^4 \sec^2\theta - 1$)

where A and A´ are the vertices of the hyperbola.

Solution :

The equation of the normal at (a secθ, b tanθ) to the given hyperbola is

$ax \cos\theta + by \cot\theta = (a^2 + b^2)$

This meets the transverse axis i.e. x-axis at G. So, the coordinates of G are $\left(\frac{a^2 + b^2}{a} \sec\theta, 0\right)$

The coordinates of the vertices A and A´ are A (a, 0) and A´ (– a, 0) respectively

$$\therefore \quad AG.A'G = \left(-a + \frac{a^2 + b^2}{a}\sec\theta\right)\left(a + \frac{a^2 + b^2}{a}\sec\theta\right)$$

$$= (-a + a\, e^2 \sec\theta)(a + e^2 \sec\theta)$$

$$= a^2 (e^4 \sec^2\theta - 1)$$

Illustration 13

Show that the locus of the middle points of the normal chords of the rectangular hyperbola $x^2 - y^2 = a^2$ is $(y^2 - x^2)^3 = 4a^2x^2y^2$.

Solution :

If (h, k) be the mid-point of the chord of the hyperbola $x^2 - y^2 = a^2$ then its equation by $T = S_1$ is

$$hx - ky = h^2 - k^2 \qquad ...(1)$$

But since (1) is normal to the hyperbola its equation is

$$x \cos\theta + y \cot\theta = 2a \qquad ...(2)$$

Putting b = a

Comparing (1) and (2), we get

$$\frac{h}{\cos\theta} = -\frac{k}{\cot\theta} = \frac{h^2 - k^2}{2a}$$

$$\therefore \quad \sec\theta = \frac{h^2 - k^2}{2ah} \text{ and } \tan\theta = \frac{h^2 - k^2}{-2ak}$$

Put in $\sec^2\theta - \tan^2\theta = 1$

$$\therefore \quad \frac{(h^2 - k^2)^2}{4a^2}\left(\frac{1}{h^2} - \frac{1}{k^2}\right) = 1$$

Hence the locus of the mid-point (h, k) is

$$(x^2 - y^2)^2 (y^2 - x^2) = 4a^2 x^2y^2$$

or $$(y^2 - x^2)^3 = 4a^2x^2y^2$$

Illustration 14

A normal to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ meets the axes in Q and R, and lines QL and RL are drawn at right angles to the axes and meet at L. [In other words the rectangle QORL is completed].

Prove that locus of the point L is the hyperbola

$$a^2x^2 - b^2y^2 = (a^2 + b^2)^2$$

Prove that further that the locus of the middle point of QR is $4(a^2x^2 - b^2y^2) = (a^2 + b^2)^2$

Solution :

Any normal to the hyperbola is

$$ax \cos\theta + by \cot\theta = a^2 + b^2 \qquad ...(1)$$

Putting y = 0 and then x = 0, we get the points Q and R where it meets the axes as

$$Q\left(\frac{a^2 + b^2}{a}\sec\theta, 0\right) \text{ and } R\left(0, \frac{a^2 + b^2}{b}\tan\theta\right)$$

Line through Q perpondicular to x-axis is $x = \frac{a^2 + b^2}{a}\sec\theta$

Line through R perpondicular to y-axis is $y = \frac{a^2 + b^2}{b}\tan\theta$

Both these lines meet at the point L and in order to find its locus, we have to eliminate θ between their equations by the help of relation $\sec^2\theta - \tan^2\theta = 1$.

or $$\frac{a^2 x^2}{(a^2 + b^2)} - \frac{b^2 y^2}{(a^2 + b^2)^2} = 1$$

or $$a^2x^2 - b^2y^2 = (a^2 + b^2)^2$$

2nd part : if (h, k) be the mid-point of QR, then

$$2h = \frac{a^2 + b^2}{a}\sec\theta, 2k = \frac{a^2 + b^2}{b}\tan\theta$$

$$4(a^2h^2 - b^2k^2) = (a^2 + b^2)^2 (\sec^2\theta - \tan^2\theta)$$

or $$(a^2x^2 - b^2y^2) = (a^2 + b^2)^2$$

is the required locus.

Chord with A Given Mid-Point

The equation of chord of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ with P (x_1, y_1) as its middle point is given by $\mathbf{T = S_1}$ where $T \equiv \frac{xx_1}{a^2} - \frac{yy_1}{b^2} - 1$ and $S_1 \equiv \frac{x_1^2}{a^2} - \frac{y_1^2}{b^2} - 1$

Chord of Contact

The equation of chord of contact of tangent drawn from a point P (x_1, y_1) to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ is T = 0 where $T \equiv \frac{xx_1}{a^2} - \frac{yy_1}{b^2} - 1$

The equation of the chord joining the points $P \equiv (a \sec\theta_1, b \tan\theta_1)$ and $Q \equiv (a \sec\theta_2, b \tan\theta_2)$ is

$$\frac{x}{a}\cos\left(\frac{\theta_1 - \theta_2}{2}\right) - \frac{y}{b}\sin\left(\frac{\theta_1 + \theta_2}{2}\right) = \cos\left(\frac{\theta_1 + \theta_2}{2}\right) \text{ or } \begin{vmatrix} x & y & 1 \\ a\sec\theta_1 & b\tan\theta_1 & 1 \\ a\sec\theta_2 & b\tan\theta_2 & 1 \end{vmatrix} = 0$$

Equation of A Diameter of A Hyperbola

The equation of the diameter bisecting chords of slope of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ is $y = \frac{b^2}{a^2 m}$

Conjugate Diameters

Two diameters of a hyperbola are said to be conjugate diameters if each bisects the chord parallel to the other. If m_1 and m_2 be the slopes of the conjugate diameters of a hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$, then $m_1 m_2 = \frac{b^2}{a^2}$

Illustration 15

Chords of the circle $x^2 + y^2 = a^2$ touch the hyperbola $x^2/a^2 - y^2/b^2 = 1$. Prove that locus of their middle point is the curve $(x^2 + y^2) = a^2x^2 - b^2y^2$.

Solution :

Let (h, k) be the mid-point of the chord of the circle $x^2 + y^2 = a^2$, so that its equation by $T = S_1$ is

$$hx + ky = h^2 + k^2$$

or $\quad y = -\dfrac{h}{k}x + \dfrac{h^2 + k^2}{k}$ i.e. of the form

$y = mx + c$

It will touch the hyperbola if $c^2 = a^2m^2 - b^2$

$$\therefore \quad \left(\frac{h^2 + k^2}{k}\right) = a^2\left(-\frac{h}{k}\right)^2 - b^2$$

or $\quad (h^2 + k^2)^2 = a^2h^2 - b^2k^2$

Generalising, the locus of mid-point (h, k) is

$(x^2 + y^2)^2 = a^2x^2 - b^2y^2$

Illustration 16

Prove that the locus of the middle points of the chords of the hyperbola $\dfrac{x^2}{a^2} - \dfrac{y^2}{b^2} = 1$ which pass through a fixed point (α, β) is a hyperbola whose centre is (α/2, β/2).

Solution :

Equation of the chord of the hyperbola whose mid-point is (h, k) by $T = S_1$ is

$$\frac{hx}{a^2} - \frac{ky}{b^2} = \frac{h^2}{a^2} - \frac{k^2}{b^2}$$

It passes through the point (α, β)

$$\therefore \quad \frac{h\alpha}{a^2} - \frac{k\beta}{b^2} = \frac{h^2}{a^2} - \frac{k^2}{b^2}$$

$$\therefore \quad \text{Locus of (h, k) is } \frac{x^2 - \alpha x}{a^2} - \frac{y^2 - \beta y}{b^2} = 0$$

$$\text{or} \quad \frac{(x - \alpha/2)^2}{a^2} - \frac{(y - \beta/2)^2}{b^2} = \frac{1}{4}\left(\frac{\alpha^2}{a^2} - \frac{\beta^2}{b^2}\right) = k^2,$$

Above equation represents a hyperbola whose centre is (α/2, β/2).

Illustration 17

From points on circle $x^2 + y^2 = a^2$ tangents are drawn to the hyperbola $x^2 - y^2 = a^2$. Prove that the locus of the middle points of the chords of contact is the curve $(x^2 - y^2) = a^2 (x^2 + y^2)$.

Solution :

Any point on the curve $x^2 + y^2 = a^2$ is $(a\cos\theta, a\sin\theta)$. Chord of contact of this point w.r.t. the hyperbola $x^2 - y^2 = a^2$ is

$$x(a\cos\theta) - y(a\sin\theta) = a^2$$

or $\quad x\cos\theta - y\sin\theta = a \qquad ...(1)$

If its mid-point be (h, k), then it is same as $T = S_1$.

or $\quad hx - ky = h^2 - k^2 \qquad ...(2)$

Comparing (1) and (2), we get

$$\frac{\cos\theta}{h} = \frac{\sin\theta}{k} = \frac{a}{h^2 - k^2}$$

But $\cos^2\theta + \sin^2\theta = 1$

$$\therefore \quad \left(\frac{ah}{h^2 - k^2}\right)^2 + \left(\frac{ak}{h^2 - k^2}\right)^2 = 1$$

Hence the locus of the mid points (h, k) is

$$a^2(x^2 + y^2) = (x^2 - y^2)^2$$

Illustration 18

For any real t $x = \dfrac{e^t + e^{-t}}{2}$, $y = \dfrac{e^t - e^{-t}}{2}$ is a point on the hyperbola $x^2 - y^2 = 1$. Show that the area bounded by the hyperbola and the lines joining its centre to the points corresponding to t_1 and $-t_1$ is t_1.

Solution :

Substituting the point in the equation of hyperbola, we get

$$\frac{1}{4}[(e^t + e^{-t})^2 - (e^t - e^{-t})^2] = \frac{1}{4}4e^te^{-t} = 1$$

Above is true for all values of t. Hence the point lies on the hyperbola $x^2 - y^2 = 1$.

Let P (x_1, y_1) be the point corresponding to t_1 then

$$x_1 = \frac{1}{2}(e^t + e^{-t}),\ y_1 = \frac{1}{2}(e^t - e^{-t})$$

If Q (x_2, y_2) be the point corresponding to $-t_1$ then

$$x_2 = \frac{1}{2}(e^{-t} + e^t) = x_1,$$

$$y_2 = \frac{1}{2}(e^{-t} - e^t) = y_1$$

$\therefore$ The point Q is (x_1, y_1). Hence PQ is a double ordinate.

The vertex A (1, 0) corresponds to t = 0

$$\Delta CPQ = \left|\frac{1}{2}(x_1 y_2 - x_2 y_1)\right| = \left|\frac{1}{2}(-x_1 y_1 - x_1 y_1)\right|$$

$$= |-x_1 y_1| = x_1 y_1$$

$$\text{Area of APMQA} = 2 \text{ area APMA} = 2\int_0^{t_1} y\frac{dx}{dt}dt$$

$$= 2\int_0^{t_1} \frac{1}{2}(e^t - e^{-t}).\frac{1}{2}(e^t - e^{-t})\,dt$$

$$= \frac{1}{2}\int_0^{t_1}(e^{2t} + e^{-2t} - 2)\,dt$$

$$= \frac{1}{2}\left[\frac{e^{2t}}{2} - \frac{e^{-2t}}{2} - 2t\right]_0^{t_1}$$

$$= \frac{1}{2}\left[\frac{e^{2t_1} - e^{-2t_1}}{2} - 2t_1\right]$$

$$= \frac{1}{2}(e^{t_1} + e^{-t_1}).\frac{1}{2}(e^{t_2} - e^{-t_1}) - t_1$$

$$= x_1 y_1 - t_1$$

Hence the required shaded area is

$$(x_1 y_1) - (x_1 y_1 - t_1) = t_1$$

Alternative : Required area $= 2\left[\Delta - \int_1^{x_1} y\,dx\right]$ ...(1)

$$\Delta = \frac{1}{2}x_1 y_1 \qquad \text{and} \qquad \int_1^{x_1} y\,dx = \int_1^{x_1}\sqrt{(x^2 - 1)}\,dx$$

$$= \left[\frac{x}{2}\sqrt{x^2 - 1} - \frac{1}{2}\log\left\{x + \sqrt{x^2 - 1}\right\}\right]_1^{x_1} = \frac{x_1}{2}\sqrt{x_1^2 - 1} - \frac{1}{2}\log\left\{x_1 + \sqrt{x_1^2 - 1}\right\} - 0$$

$$= \frac{1}{2}x_1 y_1 - \frac{1}{2}\log(x_1 + y_1) = \Delta - \frac{1}{2}\log e^{t_1} = \Delta - \frac{1}{2}t_1$$

Putting in (1), we get

$$\text{Area} = 2D - 2\Delta + 2.\frac{1}{2}t_1 = t_1$$

Pole And Polar

The polar of a point P (x_1, y_1) w.r.t. the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ is T = 0, where $T \equiv \frac{xx_1}{a^2} - \frac{yy_1}{b^2} - 1$

(i) Pole of a given line lx + my + n = 0 w.r.t. the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ is $\left(\frac{-a^2l}{n}, \frac{-b^2m}{n}\right)$

(ii) Polar of the focus is the directrix.

(iii) Any tangent is the polar of its point of contact.

(iv) If the polar of P (x_1, y_1) passes through Q (x_2, y_2) then the polar of Q will pass through P and such points are said to the conjugate points.

(v) If the pole of a line lx + my + n = 0 lies on the another line l´x + m´y + n´ = 0, then the pole of the second line will lie on the first and such lines are said to be conjugate lines.

Asymptotes

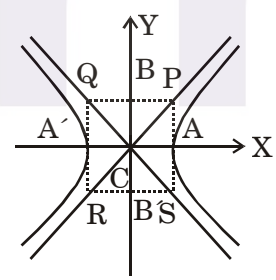
If the length of perpendicular drawn from a point on the hyperbola to a straight line tends to zero as the point on moves to infinity. The straight line is called asymptotes.

Let y = mx + c is the asymptote of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2}$.

Solving these two we get the quadratic as $(b^2 - a^2m^2)x^2 - 2a^2mcx - a^2(b^2 + c^2) = 0$

In order that y = mx + c be an asymptote, both roots of equation (1) must approach infinity.

Which are coefficient of $x^2 = 0$ & coefficient of x = 0



or $m = \pm \frac{b}{a}$ & $a^2mc = 0 \Rightarrow = 0$ ...(1)

$\therefore$ **equation of asymptote are** $\frac{x}{a} + \frac{y}{b} = 0$ **&** $\frac{x}{a} - \frac{y}{b} = 0$

Obviously angle between the asymptotes is $2\tan^{-1}(b/a)$.

If we draw lines through B, B´ parallel to the transverse axis and through A, A´ parallel to the conjugate axis, then P (a, b), Q (a, b), R (– a, – b) and S (a, – b) all lie on the asymptotes $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 0$ so asymptotes are diagonals of the rectangle PQRS. This rectangle is called associated rrectangle.

$$H\left(\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1\right), C\left(\frac{x^2}{a^2} - \frac{y^2}{b^2} = -1\right) \text{ \& } A\left(\frac{x^2}{a^2} - \frac{y^2}{b^2}\right) = 0.$$

Clearly $\rightarrow$ **C + H = 2A** {H = hyperbola, C = Conjugate hyperbola, A = Asymptotes}

Illustration 19

Find the equation of the asymptotes of the hyperbola $2x^2 - 5xy - 3y^2 - 5x - 3y - 21 = 0$.

Solution

Since the equation of the pair of asymptotes of the hyperbola differ by the hyperbola by a constant only. So, let the equation of the pair of asymptotes be

$$2x^2 - 5xy - 3y^2 - 5x - 3y + \lambda = 0 \qquad ...(i)$$

This equation represents a pair of straight lines

$$abc + 2fgh - af^2 - bg^2 - ch^2 = 0$$

Here : $a = 2$, $b = -3$, $h = -5/2$, $g = -5/2f = -3/2$ and $c = \lambda$

$$\therefore \quad -6\lambda - \frac{75}{4} - \frac{9}{2} + \frac{75}{4} - \frac{25}{4}\lambda = 0 \Rightarrow -49\lambda - 75 - 18 + 75 = 0$$

$$\Rightarrow \quad \lambda = -18/49.$$

Putting the value of λ in (i), we get the required equation

$$2x^2 - 5xy - 3y^2 - 5x - 3y - \frac{18}{49} = 0$$

Illustration 20

A series of chords of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ touch the circle on the line joining the foci as diameter. Show that the locus of the poles of these chords with respect to the hyperbola is $\frac{x^2}{a^4} + \frac{y^2}{b^4} = \frac{1}{a^2 + b^2}$.

Solution :

The foci of the hyperbola are the points S (ae, 0) and S´ (– ae, 0). Circle on SS´ as diameter is $(x - ae)(x + ae) + y^2 = 0$

or $\quad x^2 + y^2 = a^2e^2 \qquad ...(1)$

If (h, k) be the pole of the chord which touches (1), then its equation is the polar of (h, k) w.r.t. hyperbola

$$\text{i.e.} \quad \frac{hx}{a^2} - \frac{ky}{b^2} = 1 \qquad ...(2)$$

Since the line (2) touches the circle (1) therefore perpendicular from centre (0, 0) should be equal to radius ae.

$$\therefore \quad \frac{1}{\sqrt{\left(\frac{h^2}{a^4} + \frac{k^2}{b^4}\right)}} = ae$$

or $\dfrac{h^2}{a^4} + \dfrac{k^2}{b^4} = \dfrac{1}{a^2e^2} = \dfrac{1}{a^2 + b^2}$ $\therefore$ $b^2 = a^2 (e^2 - 1)$

Generalising, the locus of the pole (h, k) is the ellipse $\dfrac{x^2}{a^4} + \dfrac{y^2}{b^4} = \dfrac{1}{a^2 + b^2}$

Illustration 21

Show that the locus of poles with respect to the parabola $y^2 = 4ax$ of tangents to the hyperbola $x^2 - y^2 = a^2$ is the ellipse $4x^2 + y^2 = 4a^2$.

Solution :

Any tangent to the hyperbola $x^2 - y^2 = a^2$ is

$x \sec q - y \tan\theta = a$

$x - y \sin\theta = a \cos\theta$ ...(1)

If its pole w.r.t. parabola $y^2 = 4ax$ be (h, k) then it is same as $ky = 2a (x + h)$

or $2ax - ky = -2ah$ ...(2)

Compare (1) and (2), $\dfrac{1}{2a} = \dfrac{\sin\theta}{k} = \dfrac{\cos\theta}{-2h}$

Elimination θ by $\cos\theta + \sin^2\theta = 1$

$k^2 + 4h^2 = 4a^2$ $\therefore$ Locus is $4x^2 + y^2 = 4a^2$.

Illustration 22

A series of hyperbolas are such that the length of their transverse axis is 2a. Show that the locus of a point P on each such that its distance from the transverse axis is equal to its distance from an asymptote is the curve $(x^2 - y^2)^2 = 4x^2 (x^2 - a^2)$.

Solution :

Let the hyperbola be $\dfrac{x^2}{a^2} - \dfrac{y^2}{b^2} = 1$ and one of its asymptotes is $\dfrac{x}{a} - \dfrac{y}{b} = 0$

Here a is given but b is not known. Choose any point (h, k) on the hyperbola.

$\therefore$ $\dfrac{h^2}{a^2} - \dfrac{k^2}{b^2} = 1$ ...(1)

By given condition $k = \dfrac{\dfrac{h}{a} - \dfrac{k}{b}}{\sqrt{\left(\dfrac{1}{a^2} + \dfrac{1}{b^2}\right)}}$

or $\quad k^2\left(\frac{1}{a^2}+\frac{1}{b^2}\right)=\left(\frac{h}{a}-\frac{k}{b}\right)^2 \qquad ...(2)$

In order to find the locus of (h, k) we have to eliminate unknown b^2. Now form (2)

$$\frac{k^2}{a^2}-\frac{h^2}{a^2}=\frac{2hk}{ab}$$

$$(k^2-h^2)^2 = \frac{4a^2h^2k^2}{b^2}$$

or $\quad (k^2-h^2) = 4a^2h^2\left(\frac{h^2}{a^2}-1\right)$, by (1)

or $\quad (x^2-y^2) = 4x^2(x^2-a^2)$

is the required locus.

Illustration 23

Prove that the product of the perpendiculars from any point on the hyperbola $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$ to its asymptotes is equal to $\frac{a^2b^2}{a^2+b^2}$.

Solution :

Let (a secθ, b tanθ) be any point on the hyperbola $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$

The equations of the asymptotes of the given hyperbola are $\frac{x}{a}-\frac{y}{b}=0$ and $\frac{x}{a}+\frac{y}{b}=0$

Now, p_1 = length of the perpendicular from (a secθ, b tanθ) on $\frac{x}{a}-\frac{y}{b}=0 = \frac{\sec\theta-\tan\theta}{\sqrt{\frac{1}{a^2}+\frac{1}{b^2}}}$

and, p_2 = length of the perpendicular from on (a secθ, b tanθ) on $\frac{x}{a}+\frac{y}{b}=0 = \frac{\sec\theta+\tan\theta}{\sqrt{\frac{1}{a^2}+\frac{1}{b^2}}}$

$$\therefore\ p_1p_2 = \frac{\sec^2\theta-\tan^2\theta}{\frac{1}{a^2}+\frac{1}{b^2}}=\frac{a^2b^2}{a^2+b^2}$$

Rectangular or Equilateral Hyperbola

A hyperbola is called rectangular if its asymptotes are at right angles. The asymptotes of $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ are $y = \pm (b/a)x$ so they are perpendicular if $-b^2/a^2 = -1$ i.e., $b^2 = a^2$, i.e., $a = b$.

Hence equation of a rectangular hyperbola can be written as $x^2 - y^2 = a^2$

(i) $a^2 = a^2 (e^2 - 1)$ gives $e^2 = 2$ i.e. $e = \sqrt{2}$.

(ii) Asymptotes are $y = \pm x$

(iii) Rotating the axes by an angle $-\pi/4$ about the same origin, equation of the rectangular hyperbola $x^2 - y^2 = a^2$ is reduced to $xy = a^2/2$ or $xy = c^2$, $(c^2 = a^2/2)$. In $xy = c^2$, asymptotes are coordinate axes.

(iv) Rectangular hyperbola is also called equilateral hyperbola.

Rectangular Hyperbola referred to its asymptotes as axis of coordinates :

(i) Equation is $xy = c^2$ with parametric representation $x = ct$, $y = \frac{c}{t}$, $t \in R \sim (0)$.

(ii) Equation of chord joining the points P (t_1) & (Q (t_2), $x + t_1t_2y = c\ (t_1t_2)$

(iii) Equation of the tangent at P (x_1, y_1) is $\frac{x}{x_1} + \frac{y}{y_1} = 2$ and at P(t) is $\frac{x}{t} + ty = 2c$.

(iv) Chord with a given middle point as (h, k) is $kx + hy = 2hk$.

(v) Equation of the normal at P(t) is $x\,t^3 - yt = c\ (t^4 - 1)$

(vi) Vertex of this hyperbola is (c, c) and (– c, – c); focus is $(\sqrt{2}\,c, \sqrt{2}\,c)$ and $(-\sqrt{2}\,c, -\sqrt{2}\,c)$, the derectrices are $x + y = \pm\sqrt{2}\,c$ and l (L.R.) = $2\sqrt{2}\,c$ = T.A. = C.A.

Illustration 24

If the normal at the point 't_1' to the rectangular hyperbola $xy = c^2$ meets it again at the point 't_2', prove that $t_2 = -1/t_1^3$.

Solution :

The equation of the normal at $(ct_1, c/t_1)$ to the hyperbola $xy = c^2$ is $xt_1^3 - yt_1 - ct_1^4 + c = 0$

If this passes through $\left(ct_2, \frac{c}{t_2}\right)$, then

$$ct_2t_1^3 - \frac{c}{t_2}t_1 - ct_1^4 + c = 0 \quad \Rightarrow \quad t_2^2t_1^3 - t_1 - t_1^4t_2 + t_2 = 0$$

$$\Rightarrow \quad t_1^3t_2(t_2 - t_1) + (t_2 - t_1) = 0 \Rightarrow \quad (t_1^3t_2 + 1)(t_2 - t_1) = 0$$

$$\Rightarrow \quad t_1^3t_2 + 1 = 0 \qquad [\therefore\ t_2 \neq t_1]$$

$$\Rightarrow \quad t_2 = -1/t_1^3$$

Illustration 25

If a circle cuts the rectangular hyperbola $xy = 1$ **in the points** (x_r, y_r); $r = 1, 2, 3, 4$. **Prove that** $x_1x_2x_3x_4 = y_1y_2y_3y_4 = 1$

Solution :

Let the equation of the circle be

$$x^2 + y^2 + 2gx + 2fy + k = 0$$

The equation of the hyperbola is $xy = c^2$

Eliminating y from these two equations, we get

$$x^2 + \frac{1}{x^2} + 2gx + 2f\frac{1}{x} + k = 0 \qquad x^4 + 2gx^3 + kx^2 + 2fx + 1 = 0$$

This is a fourth degree equation in x giving four values of x, say x_1, x_2, x_3 and x_4.

$\therefore \quad x_1x_2x_3x_4 = 1$

Corresponding to every value of x, there is a value of y given by $xy = 1$.

$$\therefore \quad y_1y_2y_3y_4 = \frac{1}{x_1x_2x_3x_4} = 1$$

Illustration 26

A rectangular hyperbola whosoe centre is C is cut by any circle of radius r **in four points P, Q, R and S. Prove that** $CP^2 + CQ^2 + CR^2 + CS^2 = 4r^2$.

Solution :

Taking the rectangular asymptotes as the axes of reference the equation of the hyperbola and the circle are

$xy = k^2$ ...(1)

and $x^2 + y^2 + 2gx + 2fy + c = 0$ ...(2)

where $g^2 + f^2 - c = r^2$ (given) ...(3)

Eliminating v between (1) and (2), we get a biquadratic in x giving us the abscissas of the four points of intersection as

$$x^4 + 2gx^3 + cx^2 + 2f\,k^2\,x + k^4 = 0$$

Its roots are x_1, x_2, x_3 and x_4 then

$\Sigma x_1 = -2g$, $\Sigma x_1x_2 = c$

$$\therefore \quad x_1^2 + x_2^2 + x_3^2 + x_4^2 = \left(\sum x_1\right)^2 - 2\sum x_1x_2 = 4g^2 - 2c$$

Similarly $y_1^2 + y_2^2 + y_3^3 + y_4^2 = 4f^2 - 2c$

$\therefore \quad CP^2 + CQ^2 + CR^2 + CS^2$

$$= \left(x_1^2 + y_1^2\right) + (x_2^2 + y_2^2) + (x_3^3 + y_3^2) + (x_4^2 + y_4^2)$$

$$= \sum x_1^2 + \sum y_1^2 = 4g^2 - 2c + 4f^2 - 2c$$

$$= 4\,(g^2 + f^2 - c) = 4r^2$$

Illustration 27

If a circle cuts a rectangular hyperbola $xy = c^2$ in A, B, C and D and the parameters of these four points be t_1, t_2, t_3 and t_4 respectively, prove the following :

(a) $t_1\ t_2\ t_3\ t_4 = 1$, **find the value of** $\sum \frac{1}{t_1}$

(b) If H be the orthocentre of the triangle ABC, then H and D are extremities of a diameter of the rectangular hyperbola.

(c) The centre of mean position of the four points bisects the distance between the centres of the two curves.

(d) The centre of the circle through A, B and C is

$$\left[\frac{c}{2}\left(t_1 + t_2 + t_3 + \frac{1}{t_1t_2t_3}\right), \frac{c}{2}\left(\frac{1}{t_1} + \frac{1}{t_2} + \frac{1}{t_3} + t_1t_2t_3\right)\right]$$

Solution :

(a) Let the equation of the rectangular hyperbola referred to rectangular asymptotes as axes be

$xy = c^2$ or its parametric equations be $x = ct$ and $y = \frac{c}{t}$ ...(1)

and that of the circle be

$$x^2 + y^2 + 2gx + 2fy + k = 0 \quad ...(2)$$

Solving (1) and (2), we get

$$c^2t^2 + \frac{c^2}{t^2} + 2gct + 2f.\frac{c}{t} + k = 0$$

or $$c^2t^4 + 2gct^3 + kt^2 + 2fct + c^2 = 0 \quad ...(3)$$

Above equation being of fourth degree in t gives us the four parameters t_1, t_2, t_3 and t_4 of the points of intersection.

$$\therefore \quad t_1 + t_2 + t_3 + t_4 = \frac{2gc}{c^2} = -\frac{2g}{c} \quad ...(4)$$

$$t_1t_2t_3 + t_1t_2t_4 + t_3t_4t_1 + t_3t_4t_2 = -\frac{2fc}{c^2} = -\frac{2f}{c} \quad ...(5)$$

$$t_1t_2t_3t_4 = \frac{c^2}{c^2} = 1, \text{ which proves (a).} \quad ...(6)$$

Dividing (5) by (6), we get

$$\frac{1}{t_4} + \frac{1}{t_3} + \frac{1}{t_2} + \frac{1}{t_1} = -\frac{2f}{c}$$

(b) Orthocentre of the triangle ABC is

$$\left(\frac{-c}{t_1t_2t_3}, ct_1t_2t_3\right)$$

or H is the point $\left(-ct_4, \frac{-c}{t_4}\right)$,

$\therefore$ $t_1t_2t_3t_4 = 1$, by (6)

and D is the point $\left(ct_4, \frac{c}{t_4}\right)$ and clearly H and D are extremities of a diameter.

(c) The centre of mean position of the four points is

$$\left[\frac{c}{4}(t_1 + t_2 + t_3 + t_4), \frac{c}{4}\left(\frac{1}{t_1} + \frac{1}{t_2} + \frac{1}{t_3} + \frac{1}{t_4}\right)\right]$$

$$= \left[\frac{c}{4}\left(\frac{-2g}{c}\right), \frac{c}{4}\left(\frac{-2f}{c}\right)\right], \text{ from (4) and (7)}$$

or $\left(\frac{-g}{2}, \frac{-f}{2}\right)$ which is clearly the mid-point of (0, 0) and (– g, – f) i.e., of the join of the centres of the two curves.

(d) Let the circle through ABC meet the hyperbola in fourth point D; then from (4) and (7), the centre (– g, – f) is

$$\left[\frac{c}{2}(t_1 + t_2 + t_3 + t_4), \frac{c}{2}\left(\frac{1}{t_1} + \frac{1}{t_2} + \frac{1}{t_3} + \frac{1}{t_4}\right)\right] \quad ...(8)$$

Also $t_1t_2t_3t_4 = 1$ from (6)

$$\therefore \quad t_4 = \frac{1}{t_1t_2t_3} \quad ...(9)$$

Putting the value of t_4 in (8), we get the required centre in terms of the parameters of the given points A, B, C in the form as given.